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Batch 15

Algorithm

1. Read `size`.
2. Declare the variable-length array `arr[size]`.
3. Read all elements of `arr[]`.
4. Read `k`.
5. Pass the address of the array to `kth_smallest()`.
6. In `kth_smallest()`:
   - If `k <= 0` or `k > size`, display `-1`.
   - Otherwise, sort the array in ascending order using **Bubble Sort**.
   - After sorting, the kth smallest element is at index `k - 1`.
   - Display `*(arr + k - 1)`.
7. Stop.

Test Cases

Test	Array	k	Sorted Array	Expected Output
1	7 10 4 3 20 15	3	3 4 7 10 15 20	kth smallest element: 7
2	7 10 4 3 20 15	4	3 4 7 10 15 20	kth smallest element: 10
3	7 10 4 3 20 15	7	—	-1
4	4 2 2 7 5	3	2 2 4 5 7	kth smallest element: 4
5	5 1 8 3 2	1	1 2 3 5 8	kth smallest element: 1
6	5 1 8 3 2	5	1 2 3 5 8	kth smallest element: 8
7	1 2 3 4	0	—	-1

C program

```
#include <stdio.h>
```

```c
void kth_smallest(int *arr, int size, int k)
{
    int i, j, temp;

    /* Check whether k is valid */
    if (k <= 0 || k > size)
    {
        printf("-1\n");
        return;
    }

    /* Bubble Sort in ascending order */
    for (i = 0; i < size - 1; i++)
    {
        for (j = 0; j < size - i - 1; j++)
        {
            if (*(arr + j) > *(arr + j + 1))
            {
                temp = *(arr + j);
                *(arr + j) = *(arr + j + 1);
                *(arr + j + 1) = temp;
            }
        }
    }

    /* kth position corresponds to index k-1 */
    printf("kth smallest element: %d\n", *(arr + k - 1));
}

int main()
{
    int size, i, k;

    printf("Enter the size of the array: ");
    scanf("%d", &size);

    if (size <= 0)
    {
        printf("Invalid array size.\n");
        return 0;
    }

    int arr[size];

    printf("Enter %d elements: ", size);
    for (i = 0; i < size; i++)
    {
        scanf("%d", &arr[i]);
    }

    printf("Enter k: ");
    scanf("%d", &k);

    kth_smallest(arr, size, k);

    return 0;
}
```

Execution

```
Last login: Sun Oct 25 16:02:19 on ttys000
bhavya@Bhavyas-MacBook-Air ~ % nano day58.c
bhavya@Bhavyas-MacBook-Air ~ % clang day58.c -o day58
bhavya@Bhavyas-MacBook-Air ~ % ./day58
Enter the size of the array:6
Enter 6 elements: 7 10 4 3 20 15
Enter k: 3
kth smallest element: 7
bhavya@Bhavyas-MacBook-Air ~ % ./day58
Enter the size of the array: 5
Enter 5 elements: 5 1 8 3 2
Enter k: 1
kth smallest element: 1
bhavya@Bhavyas-MacBook-Air ~ %
```